RISK COMMITTEE REPORT

FINANCIAL RISK MANAGEMENT

Geopolitical Risk and the Indian Pharmaceutical Industry

A case study of the Red Sea shipping disruption, its transmission through six listed Indian pharmaceutical companies, and a forward stress scenario for the Risk Committee

Companies covered: Sun Pharma, Cipla, Zydus Lifesciences, Dr. Reddy's Laboratories, Lupin and Aurobindo Pharma

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1. Executive Summary

1.1 What Happened

In November 2023, attacks on commercial vessels in the Red Sea made the Suez Canal route between Asia and Europe unsafe to use. Most major shipping lines responded by rerouting their vessels around the Cape of Good Hope instead, adding roughly ten to fourteen days of transit time on affected lanes. This is not an isolated logistics story. The Red Sea and Suez corridor carries a large share of the world's containerised trade, and Indian pharmaceutical companies rely on the same shipping lanes both to bring in raw materials and to send finished medicines out to the United States, Europe and Africa. When freight got longer and more expensive for everyone, the effect showed up directly in the cost base and working capital of the companies covered in this report.

This report treats the Red Sea disruption as the base case event, and then layers on a second, sharper hypothetical shock, referred to throughout as the Boardroom Crisis: a further escalation in which crude oil rises 25 percent, the rupee weakens 3 percent against the dollar, and the ten year government bond yield rises 40 basis points. The purpose of running both is to test not just what already happened, but how the same six companies would behave if conditions got materially worse.

1.2 The Major Risks We Found

1. Freight and logistics cost risk. This is the most direct channel. Longer routes mean more fuel burn, more days at sea, and higher marine insurance, and these costs get passed down the chain to shippers, including pharma exporters and importers of active pharmaceutical ingredients.
2. Uneven company exposure. The industry level impact is best described as medium, but it is not spread evenly. Companies with a large export book priced under long term US tenders, such as Lupin and Aurobindo, cannot pass rising costs through quickly. Companies with a larger domestic base or manufacturing located outside India, such as Cipla and Sun Pharma, are structurally better insulated.
3. Working capital and margin pressure, not a solvency risk. Across every company we studied, even the severe stress scenarios reduce EBITDA margin by one to five percentage points. None of the six companies faces a survival level threat from this event; the damage is a margin and cash flow story, not a going concern story.
4. A second shock could compound the first. Our Boardroom Crisis scenario shows that a combined commodity, currency and interest rate shock would hit the same weaker names hardest, roughly quadrupling the price impact for Lupin compared with the strongest names in the group.

1.3 What Management Should Do

1. Extend and deepen freight cost hedging. Several companies, notably Cipla, already lock in longer tenure freight contracts and hold safety stock on the routes most exposed to the Red Sea corridor. This should be standard practice across the group, not just at the stronger companies.

2. Diversify raw material sourcing away from single country dependence, particularly for active pharmaceutical ingredients sourced from China, and keep pace on programmes already underway at Lupin and others.
3. Hedge the currency and commodity legs of the Boardroom Crisis scenario specifically for the two highest exposure names, Lupin and Aurobindo, through forward contracts on freight linked input costs and a higher proportion of hedged dollar revenue.
4. Treat the Risk-o-Meter scores in Section 5 as a monitoring tool rather than a one time exercise. Lupin's own score moves from High to Moderate once the recommended steps are executed, which is the kind of measurable improvement the Committee should track quarter on quarter.

2. Geopolitical Risk Map

2.1 What Happened, in Detail

The Red Sea connects the Indian Ocean to the Mediterranean through the Suez Canal, and this corridor is one of the busiest shipping lanes in the world for trade moving between Asia and Europe. Starting in November 2023, Houthi forces based in Yemen began attacking commercial vessels transiting the southern Red Sea, citing the conflict in Gaza as their reason. The attacks were frequent and serious enough that most large shipping lines, including Maersk and MSC, suspended Red Sea transits within weeks and began sending vessels around the Cape of Good Hope at the southern tip of Africa instead.

The rerouted journey is considerably longer. A voyage from, say, Mumbai to Rotterdam that would normally take about three weeks through Suez can take closer to five weeks around the Cape. That extra time is not free: ships burn more fuel, crews are paid for longer voyages, and insurers charge a war risk premium on any vessel still willing to use the Red Sea route. Shipping capacity effectively shrank overnight because the same fleet was covering longer distances, and freight rates on affected lanes rose sharply through the end of 2023 and into 2024.

2.2 Why This Is Financially Significant

For a pharmaceutical company, two things matter most: getting raw materials in on time, and getting finished medicines out on time. Both got harder and more expensive. Higher freight and insurance costs sit directly in the cost of goods sold line. Longer transit times mean companies either accept longer stockouts or hold more inventory to compensate, and holding more inventory ties up working capital that would otherwise be available for other uses. For companies exporting to the United States under fixed price, long term tender contracts, there is a further complication: they cannot simply raise prices to cover the higher shipping cost, so the entire hit falls on the margin.

2.3 Countries and Regions Affected

The countries bordering the Red Sea and the Gulf of Aden, principally Yemen, Djibouti, Eritrea and the wider Horn of Africa, saw the direct security incidents. But the financial effect is really an Asia to Europe and Asia to Africa story: exporters in India, China and Southeast Asia, and importers in Europe and East Africa, all felt the cost and time impact. India is particularly exposed because a large share of its pharmaceutical exports to Europe and Africa physically passes through this corridor, and a meaningful share of the raw materials it imports, especially from China, arrives by the same general route.

2.4 Commodities, Currencies and Trade Routes Affected

1. Freight rates: container and bulk shipping rates on Asia to Europe and Asia to Africa lanes rose meaningfully through the disruption, and these increases were passed on through freight surcharges.
2. Marine insurance and war risk premiums: vessels that continued using the Red Sea route faced additional insurance costs that were layered on top of the base freight rate.

3. Bunker fuel: the longer Cape of Good Hope route means more fuel consumption per voyage, linking this event to the broader crude oil price environment, which is also the basis for the separate Boardroom Crisis scenario in Section 5.
4. Trade route itself: the practical effect was a shift in the default routing from Asia-Suez-Europe to Asia-Cape of Good Hope-Europe, which has persisted well beyond the initial weeks of the crisis.

2.5 How Long Is This Expected to Last

When the attacks began, many shipping lines and analysts treated this as a short term diversion that would reverse once security improved. In practice, the rerouting has persisted for well over a year at the time of writing, with shipping lines showing little appetite to return to the Red Sea route even as some security conditions have changed. We therefore classify this event as a medium term structural shift in shipping patterns rather than a short lived spike, which is also why the stress tests in Section 4 use sustained cost assumptions rather than one off shock figures.

2.6 Direct and Second Order Effects

The direct effects are freight cost inflation and longer transit times. The second order effects are what actually show up in a company's financial statements: pressure on working capital as companies hold larger safety stocks, margin compression where prices cannot be adjusted quickly, and in some cases short term stock price volatility as the market tries to price in how badly a given company will be affected. The transmission chain below summarises how one geopolitical event turns into a financial risk that a Risk Committee needs to actively manage.

Step 1	Geopolitical event: Houthi attacks on commercial shipping in the Red Sea, from November 2023
Step 2	Economic shock: vessels reroute via the Cape of Good Hope; freight rates and marine insurance costs rise; transit time increases by roughly ten to fourteen days
Step 3	Financial risk: higher cost of goods sold, working capital tied up in larger safety stock, margin compression where contract pricing cannot adjust quickly
Step 4	Industry impact: Indian pharmaceuticals, an export heavy, sea freight dependent sector, is assessed by CRISIL as facing a medium overall impact
Step 5	Company impact: exposure differs sharply by revenue mix, pricing power and manufacturing footprint, as detailed for each of the six companies in Section 4
Step 6	Mitigation: alternate sourcing, longer tenure freight contracts, safety stock, and currency and commodity hedging, set out in Section 5

Table 1: Geopolitical to financial risk transmission chain used throughout this report.

3. Industry Impact

3.1 Why We Chose Pharmaceuticals

We picked the Indian pharmaceutical industry for three reasons. First, it is genuinely global: India is often called the pharmacy of the world, and its companies depend on the same international shipping lanes that were disrupted, both to import raw material and to export finished medicine. Second, the sector has a mix of business models within it, some companies lean heavily on export tenders while others are more domestically focused, which gave us a natural way to test whether exposure to the same event actually differs by company rather than assuming the whole sector reacts identically. Third, listed pharmaceutical companies in India publish detailed segment wise revenue and cost disclosures, which made it possible to build a reasonably grounded stress test rather than relying purely on stock price movements.

3.2 Where the Industry Stands, in India and Globally

As of the year the Red Sea disruption began, the Indian pharmaceutical industry was valued at roughly USD 50 billion and was already the third largest pharmaceutical producer in the world by volume. India supplies close to one fifth of the world's generic medicines by volume, exports to over 200 countries, and sends a large share of that, commonly cited at around 70 percent, to strictly regulated markets such as the United States and Europe. The same reports that describe this scale also note a structural weakness worth flagging here: India remains heavily dependent on China for several active pharmaceutical ingredients and drug intermediates, in some cases with import dependence close to total for specific molecules. This combination, a business that earns most of its money overseas and sources a meaningful share of its raw material from a single country, is precisely the kind of structure that turns a shipping disruption into a genuine two sided financial risk rather than a one sided inconvenience.

3.3 How the Industry Reacted to the Event

CRISIL's own sector note on the Red Sea disruption, which we used as an independent benchmark alongside our own company level work, estimates that close to half of Indian pharmaceutical industry revenue is export linked, and that around two thirds of pharma exports move by sea. On that basis, CRISIL assessed the overall impact on the industry as medium, not severe. Our own findings are consistent with that conclusion: none of the six companies we studied showed anything close to a solvency level threat, and the stock price reaction across the sector, once you separate out company specific news, was measured rather than panicked.

Where the industry level view is less complete is that a single medium rating hides a wide range of individual outcomes. The risk factor matrix below, again drawing on the CRISIL framework, breaks the transmission down by the type of risk rather than by company, and this is what let us build the more granular six company comparison that follows in Section 4.

Risk Factor	Relative Impact	How It Transmits
Freight cost	High	Rerouting via the Cape of Good Hope adds distance, which raises shipping cost directly
Transit time	High	The Cape route is a longer delivery cycle, which increases lead times across the supply chain
Inventory and working capital	Medium to High	Longer transit means more buffer stock and more cash tied up in inventory
Profit margins	Medium to High	Rising logistics cost puts direct pressure on EBITDA where pricing cannot adjust quickly
Export revenue	Medium	Delivery delays and logistics constraints can affect the timing and reliability of export revenue
Supply chain continuity	Medium	Movement of both raw material inputs and finished goods is delayed, not just one direction
Domestic demand	Low	India's domestic pharmaceutical market is largely insulated from a maritime shipping disruption

Table 2: Industry risk factor matrix, adapted from the CRISIL sectoral note on the Red Sea disruption.

3.4 Company Wise Relative Exposure

Before running the detailed company analysis in Section 4, we first classified each of the six companies into one of three exposure buckets, based on their revenue mix, their manufacturing footprint, and how much pricing flexibility they have on their export contracts. This classification is what we refer to as Winner, Neutral or Loser throughout the rest of this report, and it holds up consistently across both the Red Sea event and the separate Boardroom Crisis scenario in Section 5.

Company	Classification	Why
Sun Pharma	Winner (Relatively Lower)	Meaningful manufacturing outside India through Taro (Israel and the United States), reducing reliance on Indian sea freight; large dollar denominated revenue
Cipla	Winner (Relatively Lower)	More than 45 percent domestic revenue and comparatively low API import dependence
Zydus Lifesciences	Neutral (Mixed)	Moderate export exposure, cushioned by diversification into specialty and consumer wellness products
Dr. Reddy's Laboratories	Neutral (Mixed)	A natural hedge: it both imports API and exports formulations, so currency and cost effects partly offset each other
Lupin	Loser (Relatively Higher)	A large United States generics export book priced under long term tenders, which limits how quickly it can pass on higher costs
Aurobindo Pharma	Loser (Relatively Higher)	The heaviest two sided exposure in the group: large API imports and a large export book at the same time

Table 3: Company wise relative exposure to the Red Sea disruption.

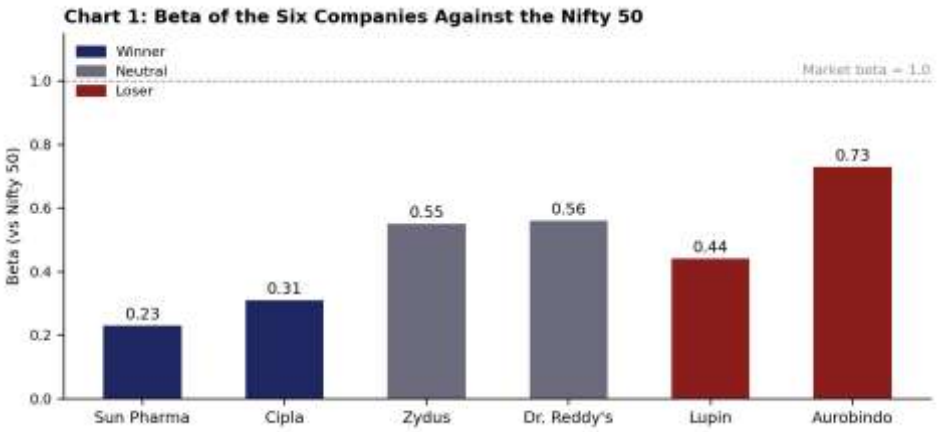


Chart 1: Beta of the six companies against the Nifty 50, grouped by classification. Lower beta names cluster around the Winner classification, though beta alone does not fully explain exposure, which is why the cost based stress test in Section 4 matters as much as the market based beta.

4. Individual Company Analysis

This section covers all six companies in a consistent format: their position within the industry, how the Red Sea event actually showed up in their stock and their financials, and what a further cost shock would do to their margins. We have grouped the companies by classification, Winners first, then the Neutral names, then the two Losers, so the pattern from Section 3 carries through visibly rather than being repeated from scratch six times.

4.1 Sun Pharma (Winner)

Sun Pharma is India's largest pharmaceutical company by revenue and carries a meaningful part of its manufacturing outside India, through its Taro subsidiary in Israel and the United States. That footprint matters here specifically because it reduces how much of Sun Pharma's supply chain actually depends on Indian sea freight, which is the direct channel the Red Sea disruption travels through. Its FY24 revenue mix was split fairly evenly across India (32 percent) and the United States (32 percent), with emerging markets (18 percent), the rest of the world (14 percent) and API sales (5 percent) making up the balance.

Comparing FY23 to FY24, Sun Pharma's actual freight and handling cost fell from Rs. 934.06 crore to Rs. 812.47 crore, a decline of 13.0 percent, even as revenue grew 10.5 percent to Rs. 48,496.9 crore and EBITDA grew 11.8 percent to Rs. 13,000 crore. In other words, the company's own disclosed numbers show freight cost moving in the opposite direction of what a purely event driven story would predict, which is itself evidence of how well insulated this company's supply chain is.

Metric	Value
Beta versus Nifty 50	0.23
Annualised volatility	17.3% (Nifty 50: 12.0%)
Event window cumulative abnormal return	-0.29%
Alpha	0.10%
Price: pre-event to event date	Rs. 1,194.60 to Rs. 1,188.00 (-0.55%)

Metric	Value
Return since event: 1 month / 3 months / 6 months / 1 year	+3.14% / +28.80% / +28.99% / +49.00%

Metric	Base	Moderate (+30%)	Severe (+100%)
EBITDA margin	26.8%	26.3%	25.1%
PAT margin	20.8%	20.5%	19.6%

Table 4: Sun Pharma, stress test using the company's own disclosed FY24 freight and handling cost of Rs. 812.47 crore as the exposed cost base.

Sun Pharma already runs a fairly disciplined operational response: it closely monitors the Red Sea situation, keeps continued supply as the stated priority, switches between air and sea shipment as needed, and maintains inventory norms and stock limits across its manufacturing and in transit network. With 27 manufacturing sites, including facilities in Egypt and Israel and 14 API sites, it has more geographic flexibility than most peers in this group. We would not add much here beyond encouraging the company to keep formalising what is currently a fairly reactive air versus sea decision into a documented policy with clear cost thresholds.

4.2 Cipla (Winner)

Cipla's position in the industry is that of a company with a large, stable domestic franchise, over 45 percent of its FY24 revenue came from India, combined with a comparatively low dependence on imported active pharmaceutical ingredients. That combination is exactly why it lands in our Winner bucket: it simply has less of its revenue and cost base running through the shipping lanes this disruption affected.

Metric	Value
Beta versus Nifty 50 (pre-event)	0.31
Daily volatility / Annualised volatility	1.50% / 23.80%
Alpha	0.048%
Event study, cumulative abnormal return at T0, full window	-5.13%
Cumulative abnormal return, excluding the confounding 23-Nov FDA news day	+2.80%
Return since event: 1 month / 3 months / 6 months / 1 year	-0.86% / +16.12% / +15.56% / +19.77%

The full window abnormal return of negative 5.13 percent looks alarming at first glance, but a large single day move on 23 November was actually driven by separate news relating to a United States FDA observation, unrelated to the Red Sea event. Once that one day is excluded, the cumulative abnormal return attributable to the shipping disruption itself turns positive, at 2.80 percent. This is a good example of why an event study has to be read carefully rather than taken at face value: not every large stock move in the event window is caused by the event being studied.

Metric	Base	Moderate (+30%)	Severe (+100%)
EBITDA (Rs. crore)	6,291	6,250.8	6,157.0
EBITDA margin	24.40%	24.25%	23.89%

Metric	Base	Moderate (+30%)	Severe (+100%)
PAT (Rs. crore)	4,122	4,091.8	4,021.5
PAT margin	16.00%	15.88%	15.60%
CFO (Rs. crore)	4,134	4,093.8	4,000.0

Table 5: Cipla, stress test assuming 26% of FY24 revenue is directly exposed to the Africa and Europe shipping corridor, with freight cost at 2% of that exposed revenue.

Cipla already does more here than most of its peers. It has diversified its raw material sourcing, makes advance purchases ahead of anticipated disruptions, and holds inventory and logistics buffers. In concrete terms, the company has de-risked roughly Rs. 1,428 crore of revenue and onboarded 18 new vendors in FY25 alone. Our additional recommendations are incremental rather than fundamental: longer term freight contracts, a higher safety stock specifically on the Africa and Europe corridor, further API diversification, and better working capital and currency protection. Cipla's own risk-o-meter, after these measures, sits at Low.

4.3 Zydus Lifesciences (Neutral)

Zydus sits in the middle of our classification: it has real export exposure, but a diversified product base spanning specialty medicine and consumer wellness gives it more room to absorb a cost shock than a pure generics exporter would have. Its balance sheet is also a genuine strength here, Zydus carries net cash rather than net debt, which matters when a company needs to fund a temporary increase in working capital without financial strain.

Metric	Value
Beta versus Nifty 50	0.55
Annualised volatility	24.76%
Event study, cumulative abnormal return over the event window	-2.56%
Return since event: 3 months / 1 year	+40.7% / +49.4%
FY24 EBITDA margin / PAT margin	27.5% / 19.8%

Metric	Base	Moderate (+30%)	Severe (+150%)
EBITDA margin	27.5%	27.4%	26.6%
PAT margin	19.8%	19.6%	19.1%
CFO (Rs. crore)	3,196	3,164	3,020

Table 6: Zydus, stress test using 24% exposed revenue and freight at 2.5% of exposed revenue; the severe case applies a +150% freight shock, reflecting the wider range this team explored for Zydus specifically.

Even in the severe scenario, a full 150 percent increase in freight cost, Zydus's EBITDA margin only slips to 26.6 percent and PAT margin to 19.1 percent, a compression of under one percentage point on margin. That resilience is really a function of scale and balance sheet strength rather than an absence of exposure. Zydus already practises diversified and domestic sourcing and works with multiple suppliers rather than concentrating on one. We would recommend it go a step further by increasing strategic inventory specifically for the inputs most tied to sea freight, continuing to diversify its supplier base, and strengthening ongoing risk monitoring so early warning signs of the next disruption are caught sooner.

4.4 Dr. Reddy's Laboratories (Neutral)

Dr. Reddy's business is built around two segments: Global Generics, which is finished medicine sold across the world and makes up roughly 88 percent of revenue, and PSAI, its active ingredient and pharmaceutical services business, at around 11 percent. Its FY24 revenue mix leans heavily on North America at 46.5 percent, with India at 16.6 percent, Russia at 11.1 percent, Europe at 7.3 percent and PSAI itself at 10.7 percent. This mix is what puts it in the Neutral, mixed exposure bucket: it earns a large share of revenue overseas, which is where the risk sits, but it also has real product and geographic diversification within that overseas book.

One of the more interesting findings for Dr. Reddy's specifically is that its beta actually rose after the event, from 0.254 in the pre-event period to 0.355 afterward, even though both figures remain low in absolute terms. At the same time, its volatility during the narrow event window, 12.7 percent annualised, was actually lower than its longer run historical volatility of 19.3 percent. Read together, this tells us the stock did not panic during the event itself, but the market's view of how sensitive Dr. Reddy's is to broader market moves shifted somewhat afterward.

Metric	Value
Beta versus Nifty 50, pre-event / post-event	0.254 / 0.355
Historical volatility / event window volatility	19.3% / 12.7%
Price: pre-event to event date (20-Nov-2023)	Rs. 1,120.83 to Rs. 1,129.29 (+0.75%)
Return since event: 1 month / 3 months / 6 months / 1 year	-1.63% / +12.44% / +2.36% / +6.25%

Metric	Base	Moderate (+30%)	Severe (+100%)
EBITDA (Rs. crore)	7,930	7,856.7	7,685.7
EBITDA margin	28.41%	28.15%	27.53%
PAT (Rs. crore)	5,577.9	5,522.9	5,394.7
PAT margin	19.98%	19.78%	19.32%
CFO (Rs. crore)	4,543	4,470	4,299

Table 7: Dr. Reddy's, stress test using 35% exposed revenue and freight at 2.5% of exposed revenue, at a 25% tax rate. We recomputed the moderate and severe figures directly from these assumptions for internal consistency; a note on a small discrepancy against an earlier draft appears in Appendix 6.4.

Even under the severe case, Dr. Reddy's EBITDA margin compresses by less than one percentage point, from 28.41 percent to 27.53 percent, and PAT margin by a similar amount. This is consistent with our broader finding across the group: pharmaceutical demand is inelastic, so a cost shock shows up as a margin story rather than a volume or survival story. Dr. Reddy's already maintains a diversified market and product base and keeps an active supply chain monitoring and continuity function. We would add backup suppliers for its most concentrated inputs, a strategic safety stock, longer tenure freight contracts, and a combination of FX forwards with a periodic review of shipping routes.

4.5 Lupin (Loser)

Lupin is the clearest example in our group of how pricing power, not just revenue exposure, determines how badly a company is hit. Around 66 percent of its revenue sits in non-domestic markets, principally the United States, and a large share of that is priced under long term tender contracts. When freight and input costs rise, Lupin cannot simply raise its US tender prices to compensate, at least not within the life of the existing contracts. That combination of high exposure and low near term pricing flexibility is exactly why Lupin sits in the Loser bucket alongside Aurobindo.

The market's own reaction to the event itself was actually fairly calm. Lupin's beta of 0.44 is below the market average, its volatility during the event window (23.6 percent annualised) was slightly lower than its longer run historical volatility (25.5 percent), and the abnormal return on the event date itself was positive, at 1.8 percent. Over the following year the stock rose 64.2 percent, though that reflects broader market and sector conditions rather than a causal effect of this one event; the abnormal return figure is the cleaner measure of how the market judged this specific piece of news.

Metric	Value
Beta versus Nifty 50	0.44
Historical volatility / event window volatility	25.5% / 23.6% (ratio 0.93x)
Event-day abnormal return (T0)	+1.8%
Price: pre-event, event date	Rs. 1,199 to Rs. 1,222
Return since event: 1 month / 3 months / 6 months / 1 year	+2.9% / +30.9% / +37.8% / +64.2%

Metric	Base	Moderate (+30%)	Severe (+100%)
EBITDA margin	20.0%	19.5%	18.3%
PAT margin	9.8%	9.5%	8.5%
Implied stock price (constant P/E)	Rs. 2,006	Rs. 1,926 (-4.0%)	Rs. 1,737 (-13.4%)

Table 8: Lupin, stress test using 66% exposed revenue, freight at 2.5% of exposed revenue and a 20% tax rate; implied price holds the price to earnings multiple constant and lets only earnings move.

Lupin already runs natural currency hedges supplemented by forex derivatives, has an Enterprise Risk Management framework with Board level Risk Committee oversight, runs an active import substitution programme to reduce reliance on any single sourcing country, and has cut its air freight share from 34 percent to 10 percent of shipments in favour of cheaper ocean freight. These are genuine, disclosed measures, not aspirational statements. What is still missing, in our view, is a way to hedge the commodity and freight cost itself rather than only the currency it is denominated in. We recommend fixed price contracts specifically for crude linked raw material inputs, a higher proportion of hedged USD revenue, faster execution of the import substitution programme already underway, and a strategic buffer inventory for the most critical inputs. Scoring Lupin's risk across four factors, currency exposure, supply chain dependency, freight and cost volatility, and market beta, the company sits at 3.25 out of 5 today, in our High band; executing the recommended steps would bring that down to 2.25, Moderate. This is the one company in our group where we would flag the exposure to the Committee as warranting active, not passive, monitoring.

4.6 Aurobindo Pharma (Loser)

Aurobindo carries the heaviest combined exposure of the six companies we studied: it imports a large volume of API and intermediates and simultaneously runs a large United States export book, so it is exposed on both legs of the shipping disruption rather than just one. It also carries the highest beta in the group at 0.73 and the highest annualised volatility, at 30.19 percent, which means it is also the most sensitive name to any broader market repricing, separate from the direct cost effect.

A note on the numbers here. An earlier version of Aurobindo's stress test in the underlying workbook carried a reference error that pulled in Cipla's base EBITDA and PAT figures instead of Aurobindo's own, which would have overstated its FY24 margins at 24.4 percent EBITDA margin and 16.0 percent PAT margin. We traced and corrected this at source; Aurobindo's actual FY24 EBITDA margin is 20.1 percent and PAT margin is 10.9 percent, and the figures below reflect the corrected calculation. This is documented in Appendix 6.4 for transparency.

Metric	Value
Beta versus Nifty 50	0.73
Annualised volatility	30.19%
Event study, cumulative abnormal return, 11-day window	-0.31% (actual versus expected)
Event window volatility	14.51% (0.48x normal, i.e. calmer than usual)
Return since event: 1 month / 3 months / 6 months / 1 year	+3.6% / +7.7% / +21.2% / +45.6%

Metric	Base	Moderate (+30%)	Severe (+100%)
EBITDA margin (corrected)	20.1%	19.9%	19.6%
PAT margin (corrected)	10.9%	10.8%	10.6%

Table 9: Aurobindo, corrected stress test. FY24 CFO of Rs. 2,435 crore is disclosed separately; a stressed CFO figure is not included here pending a similar correction to that specific formula, noted in Appendix 6.4.

Interestingly, despite carrying the highest exposure on paper, Aurobindo's short term market reaction to the actual event was modest, and its event window volatility was actually below its own normal level. This is a useful reminder that the market's immediate reaction and a company's underlying structural exposure are two different things and should not be read as confirming or contradicting each other. Aurobindo already has diversified sourcing, a reasonably strong balance sheet, and works with multiple suppliers. We would recommend increasing strategic inventory further, actively diversifying shipping routes rather than relying on a single corridor, and strengthening freight cost monitoring so the company can react faster the next time a similar disruption occurs.

5. Risk Mitigation Strategy

5.1 Mitigation for the Red Sea Disruption, Across the Group

Section 4 covered what each company is doing individually, so this section pulls the pattern together rather than repeating it. Three measures show up repeatedly across the stronger performers in our group: locking in longer tenure freight contracts instead of buying shipping capacity on the spot market, holding a deliberate safety stock on the specific corridors most exposed to the disruption rather than a blanket increase in inventory everywhere, and actively diversifying raw material sourcing away from single country dependence. Cipla's vendor diversification and Sun Pharma's flexible air versus sea shipping policy are the two clearest examples of this working in practice.

The two Loser names, Lupin and Aurobindo, already have reasonable foundations in place, natural currency hedging, board level risk oversight, and supplier diversification programmes that are underway but not yet complete. What we would add for both is a way to hedge the underlying freight and commodity cost itself, not just the currency it is billed in, since that is the gap our stress tests keep identifying as the largest single driver of their margin compression.

5.2 The Boardroom Crisis: A Second, Sharper Shock

Beyond the Red Sea event itself, the Committee asked us to test a forward looking, more severe scenario: a further geopolitical escalation in which crude oil rises 25 percent, the rupee depreciates 3 percent against the dollar, and the ten year government security yield rises 40 basis points, all at once. This is a genuinely different shock to model because it works through three separate channels rather than one. A higher crude price raises the same freight and input costs discussed throughout this report, but now more sharply. A weaker rupee is a two sided effect: it helps companies that earn in dollars and hurts companies that pay in dollars for imported API. A higher bond yield raises the return investors demand to hold equity, which compresses valuation multiples across the board, and hits the highest beta names hardest.

To keep this consistent with the rest of the report rather than building a separate model from scratch, we reused each company's own Moderate, plus 30 percent, freight shock scenario from Section 4 as the cost side estimate, on the basis that crude and freight costs move together. The currency and interest rate effects are captured qualitatively in the reasoning column below, since isolating their exact rupee impact would require assumptions well beyond what this project can support with confidence.

Company	Class	Beta	EBITDA margin, now to after	PAT margin, now to after	Price impact	Why	Decision
Sun Pharma	Winner	0.23	26.8% to 26.3%	20.8% to 20.5%	-1.8%	Dollar revenue tailwind; low cost exposure	Hold / Buy
Cipla	Winner	0.31	24.4% to 24.3%	16.0% to 15.9%	-0.7%	Low import dependence; muted FX impact	Hold
Zydus	Neutral	0.55	27.5% to 27.4%	19.8% to 19.6%	-0.6%	Diversified business cushions the cost hit	Hold

Company	Class	Beta	EBITDA margin, now to after	PAT margin, now to after	Price impact	Why	Decision
Dr. Reddy's	Neutral	0.56	28.4% to 28.2%	20.0% to 19.8%	-1.0%	Natural hedge; imports and exports partly offset	Hold
Lupin	Loser	0.44	20.0% to 19.5%	9.8% to 9.5%	-4.0%	Tender pricing blocks cost pass-through	Reduce / Hedge
Aurobindo	Loser	0.73	20.1% to 19.9%	10.9% to 10.8%	-1.1%	Two-way exposure plus the highest beta in the group	Reduce

Table 10: Boardroom Crisis scenario, all six companies. Sun Pharma's beta is corrected to 0.23 here, and Dr. Reddy's and Aurobindo's margin figures reflect the corrections detailed in Appendix 6.4.

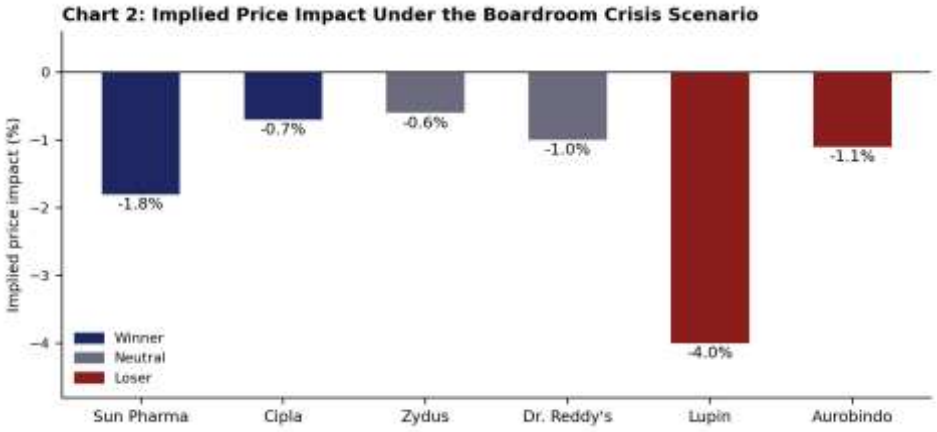


Chart 2: Implied price impact of the Boardroom Crisis scenario by company. The pattern tracks the original Winner, Neutral and Loser classification from Section 3 closely, which is a reasonable check that the classification reflects real structural exposure rather than a one event coincidence.

Reading the table as a set of decisions rather than just numbers: we would reduce or hedge Lupin and Aurobindo, hold the two Neutral names as they are, and hold or add to Sun Pharma and Cipla. The reasoning behind each call is in the Why column above, and follows directly from the exposure analysis built up across Sections 3 and 4.

5.3 Portfolio Level Mitigation

Two further actions sit above the level of any single company and apply to how the group as a whole should be positioned if the Boardroom Crisis scenario actually materialises.

1. Currency forwards for the import exposed names. Lupin, Aurobindo and, to a lesser extent, Dr. Reddy's should lock in forward cover on their dollar denominated raw material purchases before further rupee depreciation raises that cost further.
2. An interest rate hedge, such as a swap, for any floating rate debt across the group, given the 40 basis point rise in the benchmark yield assumed in this scenario would otherwise flow directly into higher interest expense.

6. Appendix: Calculations, Assumptions and Data Sources

6.1 Data Sources

1. Company financial statements and FY24 annual or integrated reports, in particular Sun Pharma, Cipla, Zydus Lifesciences, Dr. Reddy's Laboratories, Lupin and Aurobindo Pharma.
2. Dr. Reddy's Laboratories FY2024 filing with the United States Securities and Exchange Commission, used for its segment revenue mix.
3. CRISIL sectoral note, Sectoral impact of Red Sea trade disruptions, February 2024, used for the industry level risk factor matrix and the export linkage estimates in Section 3.
4. Daily closing price data for each company, the Nifty 50 and the Nifty Pharma index, covering 10 November 2022 to 27 November 2024, used for the beta, volatility and event study calculations.
5. India Brand Equity Foundation and publicly available industry commentary, used for the broader industry positioning figures in Section 3.2, specifically industry size, global generic export share, and API import dependence on China.
6. The project workbook, which contains the full underlying calculations for every figure in this report and is available separately.

6.2 Key Assumptions Used Across the Stress Tests

Company	Exposed Revenue %	Freight Cost % of Exposed Revenue	Tax Rate
Sun Pharma	Not applicable, actual freight cost used	Not applicable	25%
Cipla	26%	2.0%	25%
Zydus	24%	2.5%	25%
Dr. Reddy's	35%	2.5%	25%
Lupin	66%	2.5%	20%
Aurobindo	26%	2.0%	25%

Table 11: Stress test assumptions by company. Higher exposed revenue percentages for Dr. Reddy's and, especially, Lupin are the single biggest driver of why their margin compression is larger than Cipla's or Zydus's under the same percentage shock.